Selected advances from Deep Reinforcement Learning

Papers and new ideas

Dueling networks - 2015

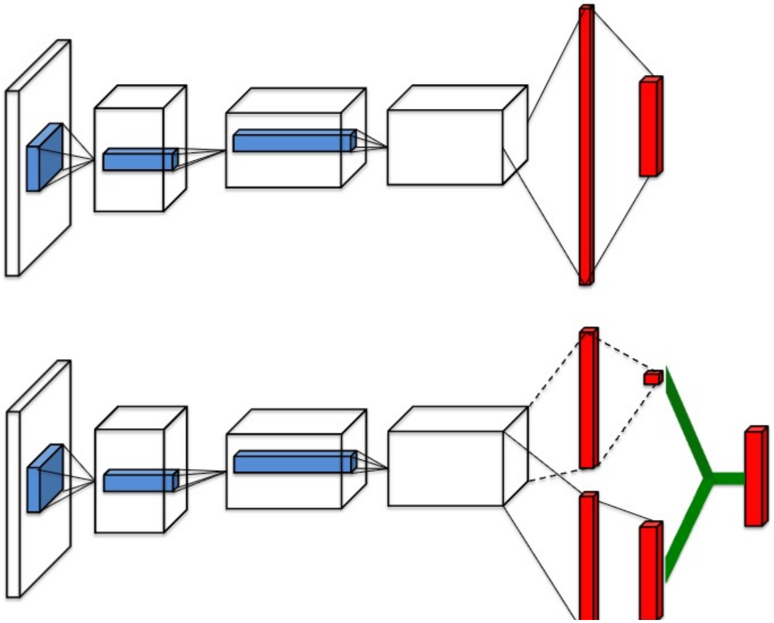


Figure 1. A popular single stream Q-network (**top**) and the dueling Q-network (**bottom**). The dueling network has two streams to separately estimate (scalar) state-value and the advantages for each action; the green output module implements equation (9) to combine them. Both networks output Q-values for each action.

$$Q(s, a; \theta, \alpha, \beta) = V(s; \theta, \beta) + \left(A(s, a; \theta, \alpha) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(s, a'; \theta, \alpha) \right)$$

$$A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s)$$

https://arxiv.org/pdf/1511.06581

Rainbow DQNs – combining improvements

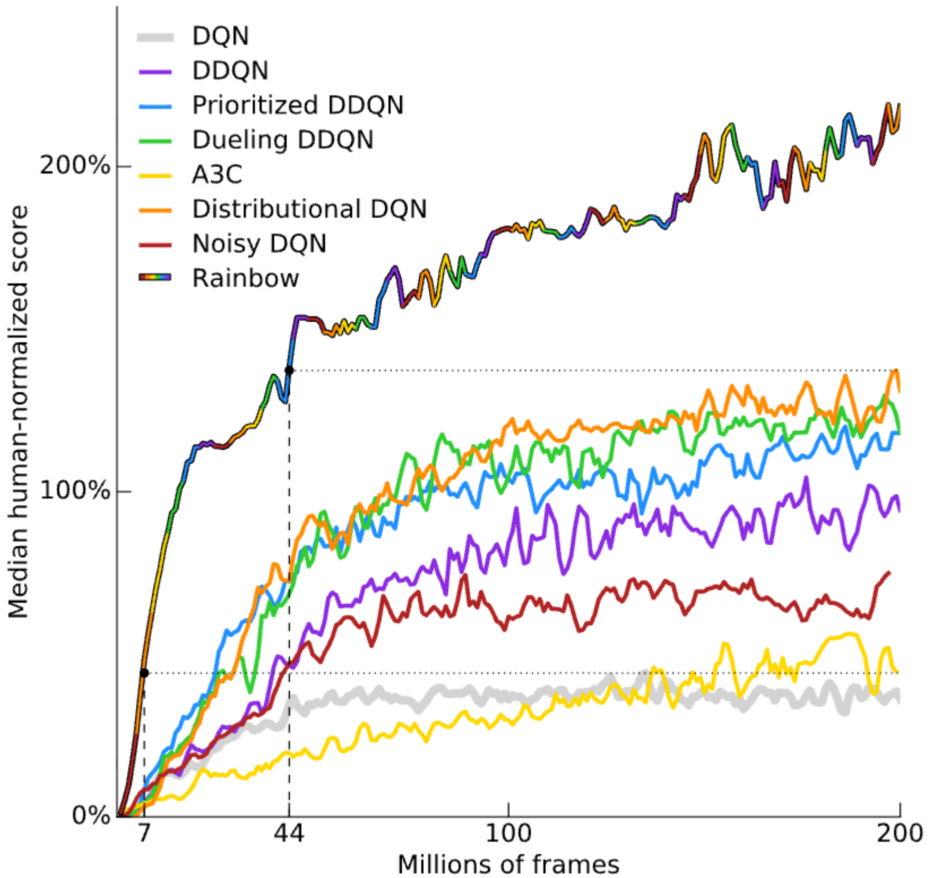


Figure 1: **Median human-normalized performance** across 57 Atari games. We compare our integrated agent (rainbow-colored) to DQN (grey) and six published baselines. Note that we match DQN's best performance after 7M frames, surpass any baseline within 44M frames, and reach substantially improved final performance. Curves are smoothed with a moving average over 5 points.

https://arxiv.org/pdf/1710.02298

Conservative Q-Learning for Offline Reinforcement Learning

Aviral Kumar[1], Aurick Zhou[1], George Tucker[2], Sergey Levine[1,2]

Major advances in offline RL

https://arxiv.org/pdf/2006.04779

Abstract

Effectively leveraging large, previously collected datasets in reinforcement learning (RL) is a key challenge for large-scale real-world applications. Offline RL algorithms promise to learn effective policies from previously-collected, static datasets without further interaction. However, in practice, offline RL presents a major challenge, and standard off-policy RL methods can fail due to overestimation of values induced by the distributional shift between the dataset and the learned policy, especially when training on complex and multi-modal data distributions. In this paper, we propose *conservative Q-learning (CQL)*, which aims to address these limitations by learning a conservative Q-function such that the expected value of a policy under this Q-function lower-bounds its true value. We theoretically show that CQL produces a lower bound on the value of the current policy and that it can be incorporated into a policy learning procedure with theoretical improvement guarantees. In practice, CQL augments the standard Bellman error objective with a simple Q-value regularizer which is straightforward to implement on top of existing deep Q-learning and actor-critic implementations. On both discrete and continuous control domains, we show that CQL substantially outperforms existing offline RL methods, often learning policies that attain 2-5 times higher final return, especially when learning from complex and multi-modal data distributions.

Offline Reinforcement Learning with Implicit Q-Learning

Ilya Kostrikov, Ashvin Nair & Sergey Levine

Major advances in offline RL

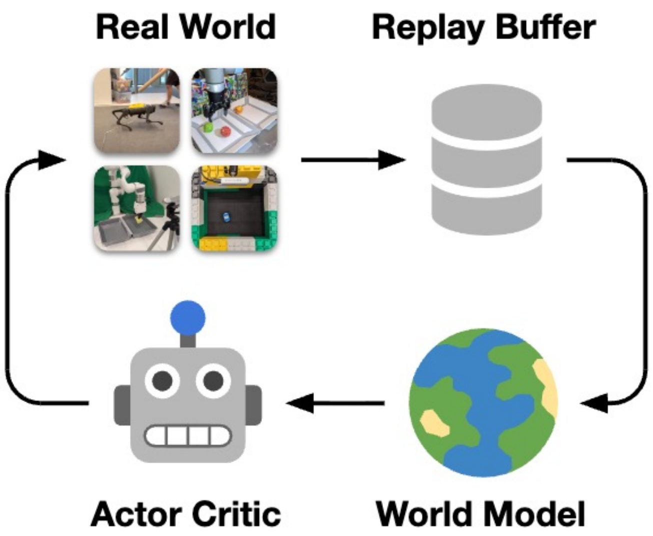
https://arxiv.org/pdf/2006.04779

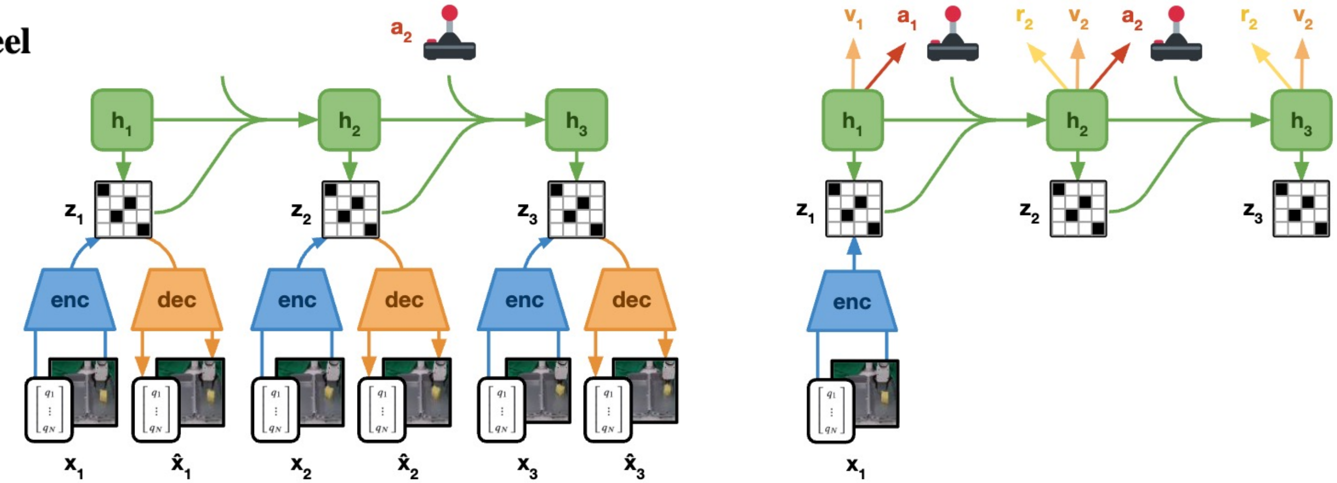
actions to be in-distribution, or else regularize their values. We propose a new offline RL method that never needs to evaluate actions outside of the dataset, but still enables the learned policy to improve substantially over the best behavior in the data through generalization. The main insight in our work is that, instead of evaluating unseen actions from the latest policy, we can approximate the policy improvement step implicitly by treating the state value function as a random variable, with randomness determined by the action (while still integrating over the dynamics to avoid excessive optimism), and then taking a state conditional upper expectile of this random variable to estimate the value of the best actions in that state. This leverages the generalization capacity of the function approximator to estimate the value of the best available action at a given state without ever directly querying a Q-function with this unseen action. Our algorithm alternates between fitting this upper expectile value function and backing it up into a Q-function, without any explicit policy. Then, we extract the policy via advantage-weighted behavioral cloning, which also avoids querying out-of-sample actions. We dub our method implicit Q-learning (IQL). IQL is easy to implement, computationally efficient, and only requires fitting an additional critic with an asymmetric L2 loss.[1] IQL demonstrates the state-of-the-art performance on D4RL, a standard benchmark for offline reinforcement learning. We also demonstrate that IQL achieves strong performance fine-tuning using online interaction after offline initialization.

DayDreamer: World Models for Physical Robot Learning

Philipp Wu* Alejandro Escontrela* Danijar Hafner*

Ken Goldberg Pieter Abbeel





(a) World Model Learning (b) Behavior Learning

Figure 3: **Neural Network Training** We leverage the Dreamer algorithm (Hafner et al., 2019; 2020) for fast robot learning in real world. Dreamer consists of two neural network components. **Left:** The world model follows the structure of a deep Kalman filter that is trained on subsequences drawn from the replay buffer. The **encoder** fuses all sensory modalities into discrete codes. The decoder reconstructions the inputs from the codes, providing a rich learning signal and enabling human inspection of model predictions. A **recurrent state-space model** (RSSM) is trained to predict future codes given actions, without observing intermediate inputs. **Right:** The world model enables massively parallel policy optimization from imagined rollouts in the compact latent space using a large batch size, without having to reconstruct sensory inputs. Dreamer trains a **policy network** and **value network** from the imagined rollouts and a learned **reward function**.

MBMF algorithms (hybrid)

- Use a learned dynamics model to *guide or accelerate* a model-free learner
- Typically alternates or combines:
  - learning a dynamics model from data
  - using that model for planning or generating synthetic rollouts
  - training a model-free policy/ value function with both real and model-generated data

MBMF: Model-Based Priors for Model-Free Reinforcement Learning

Somil Bansal Roberto Calandra Kurtland Chua Sergey Levine Claire Tomlin

Abstract: Reinforcement Learning is divided in two main paradigms: model-free and model-based. Each of these two paradigms has strengths and limitations, and has been successfully applied to real world domains that are appropriate to its corresponding strengths. In this paper, we present a new approach aimed at bridging the gap between these two paradigms that is at the same time data-efficient and cost-savvy. We do so by learning a probabilistic dynamics model and leveraging it as a prior for the intertwined model-free optimization. As a result, our approach can exploit the generality and structure of the dynamics model, but is also capable of ignoring its inevitable inaccuracies, by directly incorporating the evidence provided by the direct observation of the cost. Preliminary results demonstrate that our approach outperforms purely model-based and model-free approaches, as well as the approach of simply switching from a model-based to a model-free setting.

MBMF algorithms

Neural Network Dynamics for Model-Based Deep Reinforcement Learning with Model-Free Fine-Tuning

Anusha Nagabandi, Gregory Kahn, Ronald S. Fearing, Sergey Levine
University of California, Berkeley

***Abstract*— Model-free deep reinforcement learning algorithms have been shown to be capable of learning a wide range of robotic skills, but typically require a very large number of samples to achieve good performance. Model-based algorithms, in principle, can provide for much more efficient learning, but have proven difficult to extend to expressive, high-capacity models such as deep neural networks. In this work, we demonstrate that medium-sized neural network models can in fact be combined with model predictive control (MPC) to achieve excellent sample complexity in a model-based reinforcement learning algorithm, producing stable and plausible gaits to accomplish various complex locomotion tasks. We also propose using deep neural network dynamics models to initialize a model-free learner, in order to combine the sample efficiency of model-based approaches with the high task-specific performance of model-free methods. We empirically**

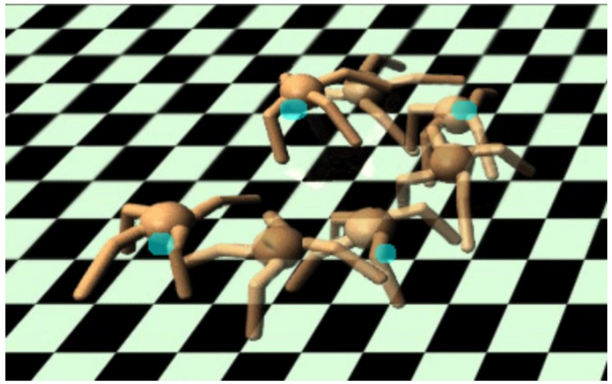


Fig. 1: Our method can learn a model that enables a simulated quadrupedal robot to autonomously discover a walking gait that follows user-defined waypoints at test time. Training for this task used 7e5 time steps, collected without any knowledge of the test-time navigation task.

MVBE - Model-Based Value Expansion

- Uses the learned model specifically to *improve value targets* rather than to generate full synthetic trajectories.
- Instead of long model rollouts, it:
  - rolls out the model for a small number of steps (k-step)
  - then bootstraps using a learned value function
- Usually plugged into off-policy algorithms:
  - DDPG
  - SAC

Model-Based Value Expansion
for Efficient Model-Free Reinforcement Learning

Vladimir Feinberg [1] **Alvin Wan** [1] **Ion Stoica** [1] **Michael I. Jordan** [1] **Joseph E. Gonzalez** [1] **Sergey Levine** [1]

Abstract

Recent model-free reinforcement learning algorithms have proposed incorporating learned dynamics models as a source of additional data with the intention of reducing sample complexity. Such methods hold the promise of incorporating imagined data coupled with a notion of model uncertainty to accelerate the learning of continuous control tasks. Unfortunately, they rely on heuristics that limit usage of the dynamics model. We present *model-based value expansion*, which controls for uncertainty in the model by only allowing imagination to fixed depth. By enabling wider use of learned dynamics models within a model-free reinforcement learning algorithm, we improve value estimation, which, in turn, reduces the sample complexity of learning.

Deep Reinforcement Learning at the Edge of the Statistical Precipice

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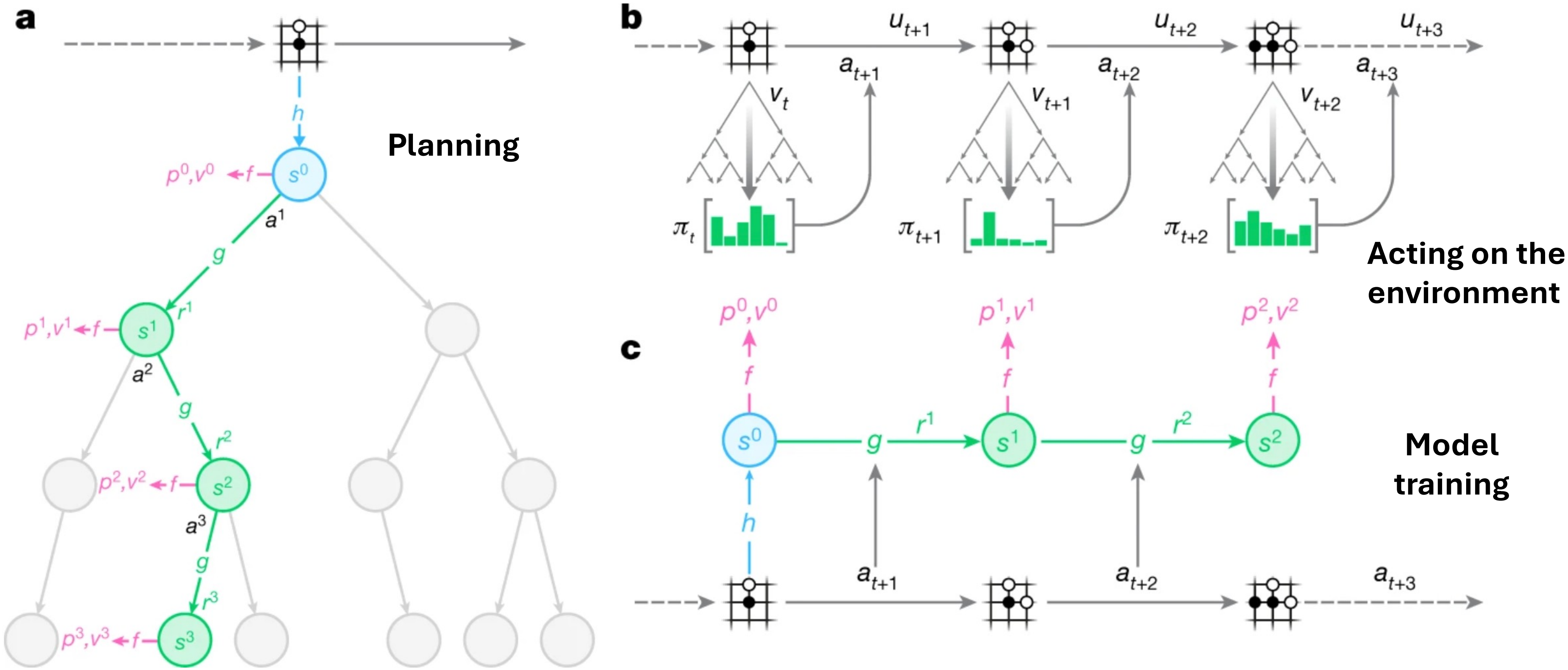
the statistical uncertainty in point estimates. In this paper, we argue that reliable evaluation in the few-run deep RL regime cannot ignore the uncertainty in results without running the risk of slowing down progress in the field. We illustrate this point using a case study on the Atari 100k benchmark, where we find substantial discrepancies between conclusions drawn from point estimates alone versus a more thorough statistical analysis. With the aim of increasing the field's confidence in reported results with *a handful of runs*, we advocate for reporting interval estimates of aggregate performance and propose performance profiles to account for the variability in results, as well as present more robust and efficient aggregate metrics, such as interquartile mean scores, to achieve small uncertainty in results. Using such statistical tools, we scrutinize performance evaluations of existing algorithms on other widely used RL benchmarks including the ALE, Procgen, and the DeepMind Control Suite, again revealing discrepancies in prior comparisons. Our findings call for a change in how we evaluate performance in deep RL, for which we present a more rigorous evaluation methodology, accompanied with an open-source library *rliable*[2], to prevent unreliable results from stagnating the field.

Mastering Atari, Go, chess and shogi by planning with a learned model

Muzero

Julian Schrittwieser, Ioannis Antonoglou, Thomas Hubert, Karen Simonyan, Laurent Sifre, Simon Schmitt, Arthur Guez, Edward Lockhart, Demis Hassabis, Thore Graepel, Timothy Lillicrap & David Silver



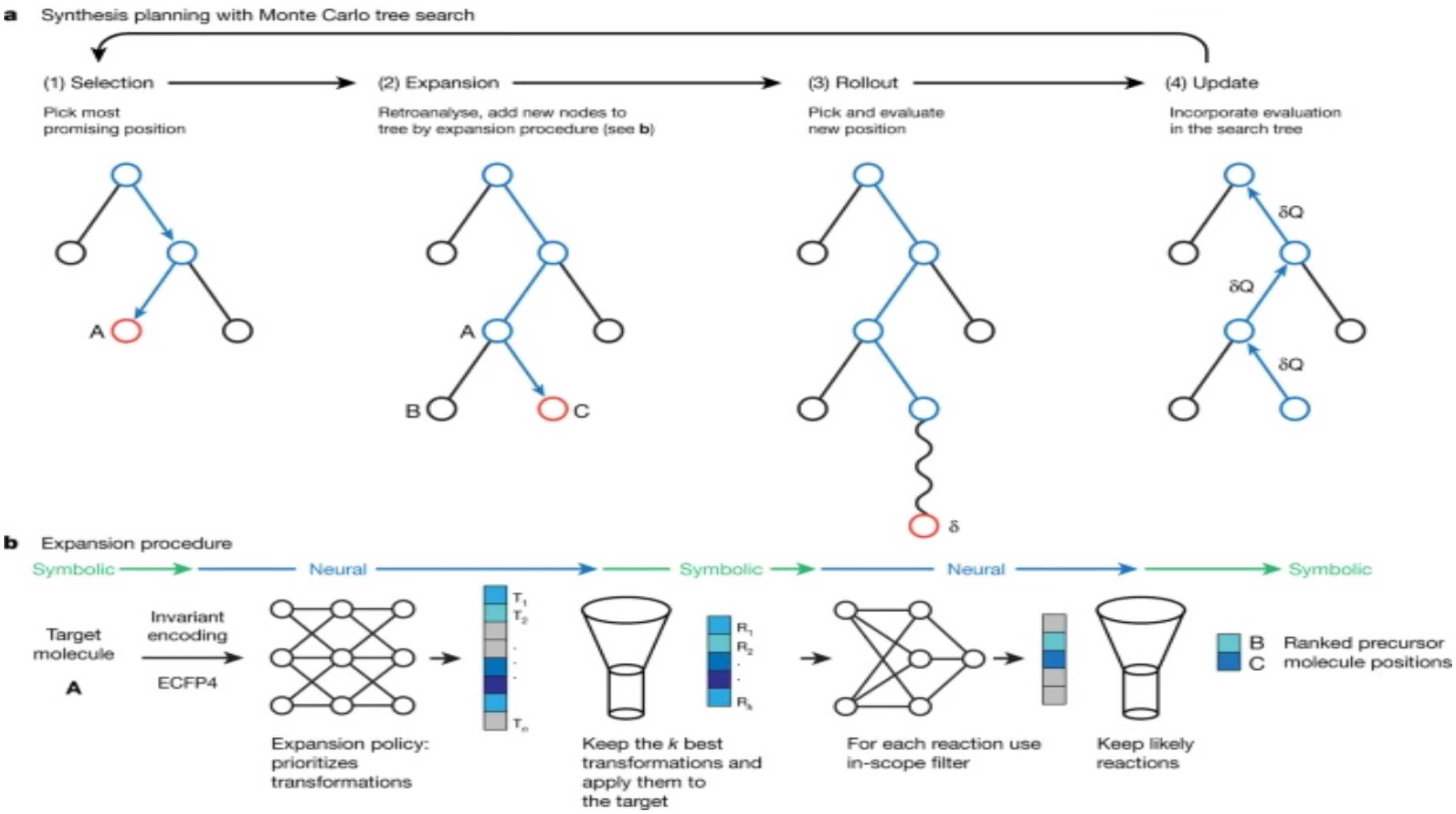
Article | Published: 29 March 2018

Planning chemical syntheses with deep neural networks and symbolic AI

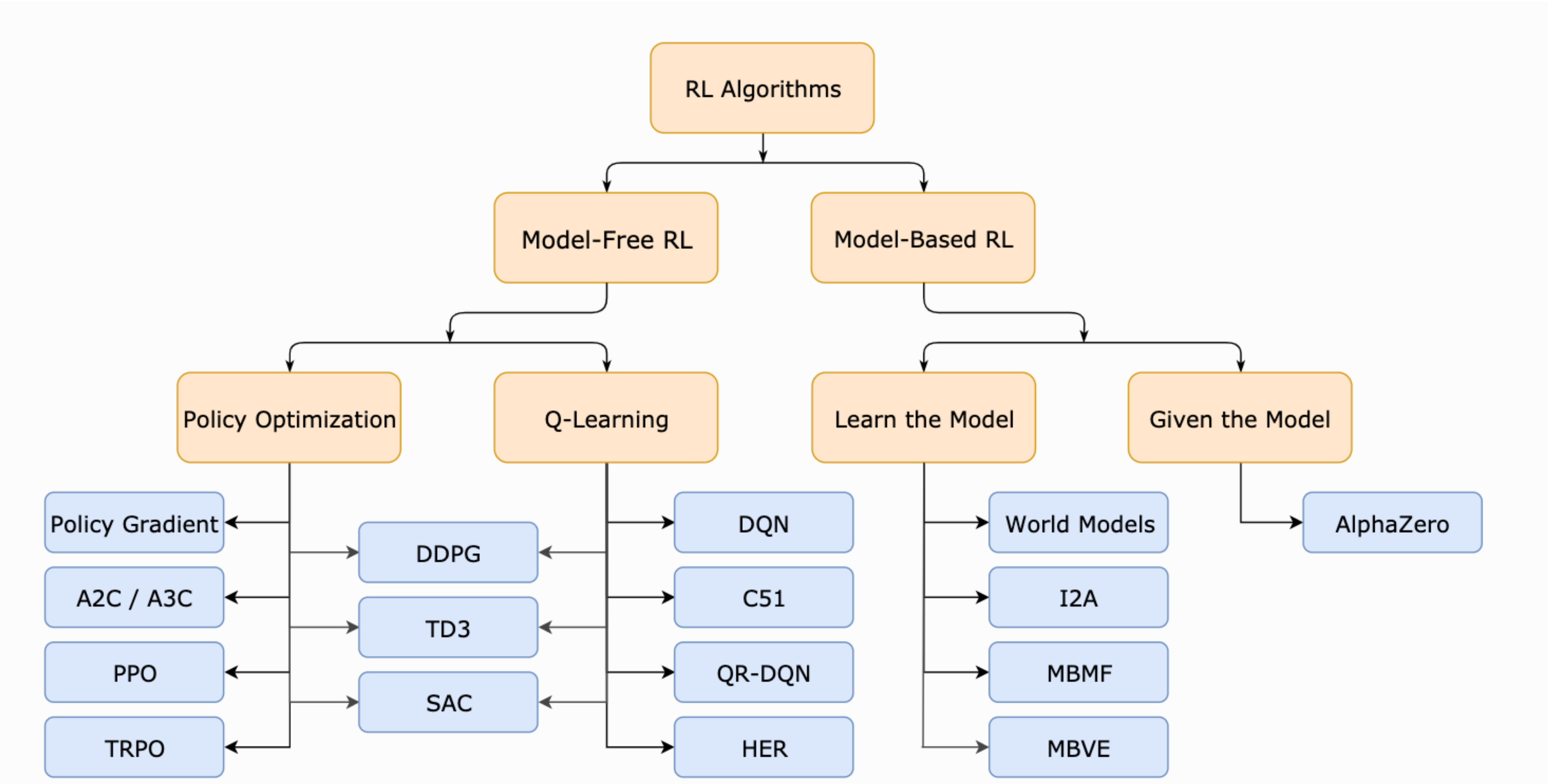
https://www.nature.com/articles/nature25978

Marwin H. S. Segler, Mike Preuss & Mark P. Waller

Nature **555**, 604–610 (2018)



Taxonomy of RL algorithms



Reward is enough

David Silver, Satinder Singh, Doina Precup, Richard S. Sutton

Abstract

In this article we hypothesise that intelligence, and its associated abilities, can be understood as subserving the maximisation of reward. Accordingly, reward is enough to drive behaviour that exhibits abilities studied in natural and artificial intelligence, including knowledge, learning, perception, social intelligence, language, generalisation and imitation. This is in contrast to the view that specialised problem formulations are needed for each ability, based on other signals or objectives. Furthermore, we suggest that agents that learn through trial and error experience to maximise reward could learn behaviour that exhibits most if not all of these abilities, and therefore that powerful reinforcement learning agents could constitute a solution to artificial general intelligence.

https://www.sciencedirect.com/science/article/pii/S0004370221000862

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Discovering state-of-the-art reinforcement learning algorithms

Junhyuk Oh, Gregory Farquhar, Iurii Kemaev, Dan A. Calian, Matteo Hessel, Luisa Zintgraf, Satinder Singh, Hado van Hasselt & David Silver

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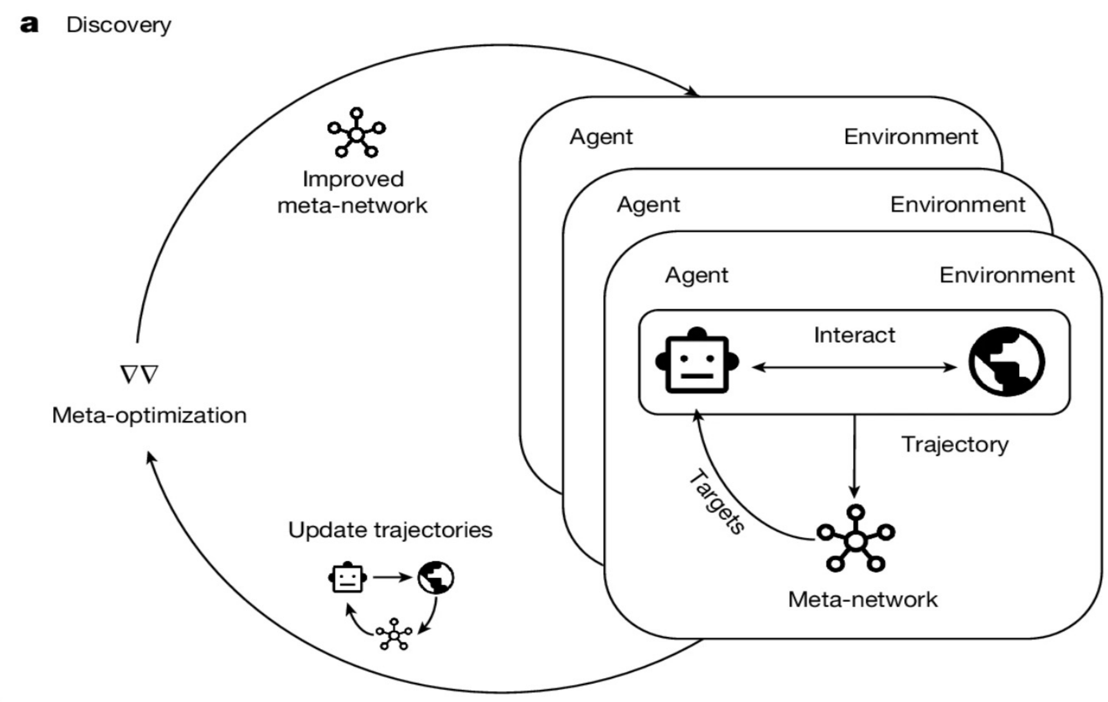

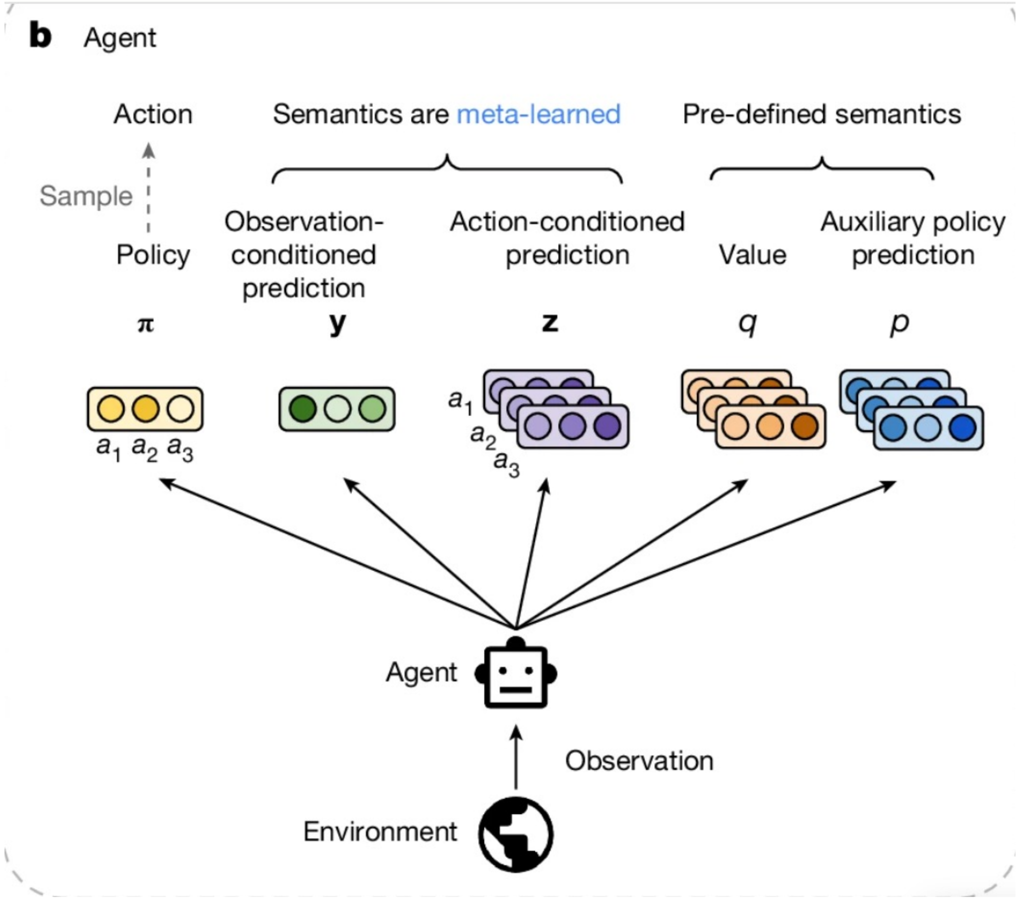

https://www.nature.com/articles/s41586-025-09761-x

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Discovering state-of-the-art reinforcement learning algorithms

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